

# Does the Prior Help when Data is Scarce? Ill-Posed Image-Space Supervision

Patient-Specific Residual INR — SLAM Ax\_T2\_2D, change\_extent=1

July 13, 2026

## Setup

We reconstruct a single follow-up slice from only a subset of its pixels, to probe the regime that actually matters for accelerated MRI: *insufficient* data. The residual (or, for baselines, the reconstruction) is supervised only on the observed pixels and evaluated on the *full* image. Four methods share the same observed pixels: **current-only INR** (no prior, the robust floor); **prior-finetune** (NeRP-style, the prior INR fine-tuned on observed pixels); **ours** (frozen prior + residual); and **ours+gate** (adds a spatial trust gate). If the prior is worth its keep, the prior-anchored methods should hold up as supervision shrinks while current-only collapses.

### A. Very sparse random supervision

A random fraction  $p$  of pixels is observed (scattered across the image), swept from 50% down to 2%. As  $p$  falls the gaps between observed pixels grow, so pure interpolation (current-only) degrades; a prior can fill the gaps with plausible anatomy.

Table 1: Very sparse random: full-image metrics vs. observed fraction  $p$ .

| $p$ | method                                  | PSNR  | SSIM  | change-gain |
|-----|-----------------------------------------|-------|-------|-------------|
| 50% | current-only INR                        | 32.11 | 0.914 | 0.977       |
|     | prior-finetune (NeRP)                   | 32.44 | 0.925 | 0.975       |
|     | <b>prior+residual (ours)</b>            | 28.44 | 0.813 | 0.883       |
|     | <b>prior+gated residual (ours+gate)</b> | 29.05 | 0.904 | 0.904       |
| 25% | current-only INR                        | 28.48 | 0.846 | 0.953       |
|     | prior-finetune (NeRP)                   | 29.15 | 0.873 | 0.942       |
|     | <b>prior+residual (ours)</b>            | 25.37 | 0.763 | 0.776       |
|     | <b>prior+gated residual (ours+gate)</b> | 25.64 | 0.839 | 0.785       |
| 10% | current-only INR                        | 24.41 | 0.766 | 0.925       |
|     | prior-finetune (NeRP)                   | 25.72 | 0.799 | 0.872       |
|     | <b>prior+residual (ours)</b>            | 22.60 | 0.695 | 0.613       |
|     | <b>prior+gated residual (ours+gate)</b> | 22.32 | 0.739 | 0.566       |
| 5%  | current-only INR                        | 22.24 | 0.692 | 0.898       |
|     | prior-finetune (NeRP)                   | 24.19 | 0.724 | 0.807       |
|     | <b>prior+residual (ours)</b>            | 21.54 | 0.679 | 0.506       |
|     | <b>prior+gated residual (ours+gate)</b> | 21.31 | 0.718 | 0.429       |
| 2%  | current-only INR                        | 20.27 | 0.565 | 0.951       |
|     | prior-finetune (NeRP)                   | 22.81 | 0.571 | 0.720       |
|     | <b>prior+residual (ours)</b>            | 20.78 | 0.619 | 0.415       |
|     | <b>prior+gated residual (ours+gate)</b> | 20.57 | 0.698 | 0.351       |

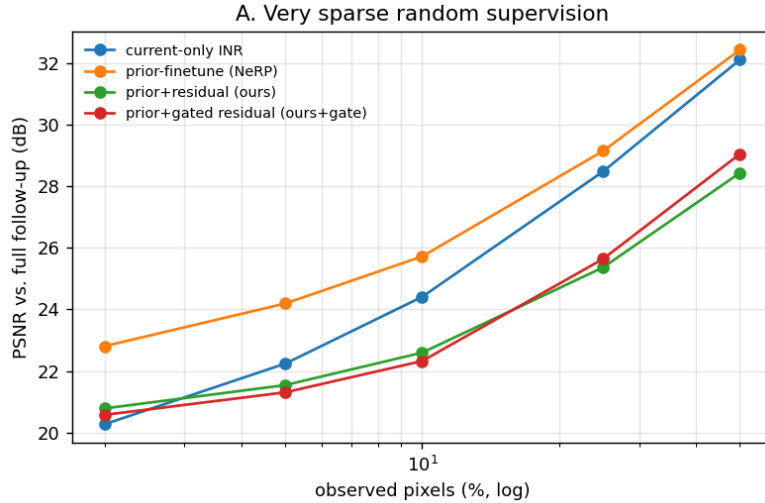


Figure 1: PSNR vs. observed fraction (log axis). A flat curve = robust to data scarcity.

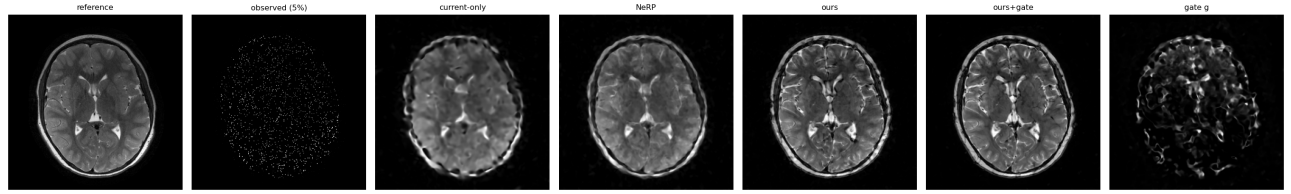


Figure 2: Reconstructions at  $p = 5\%$ : reference, observed pixels, then each method, and the gate map (white = distrust prior).

## B. Structured hole

Everything is observed *except* a central square hole of side  $h$ ; the hole has no interior samples, so it can only be filled by extrapolation or from the prior. As  $h$  grows the reconstruction inside the hole depends entirely on the prior.

Table 2: Structured hole: PSNR inside the hole (the filled region) and over the full image.

| hole $h$ | method                                  | PSNR(hole) | PSNR(full) | SSIM(full) |
|----------|-----------------------------------------|------------|------------|------------|
| 0        | current-only INR                        | 38.95      | 38.95      | 0.961      |
|          | prior-finetune (NeRP)                   | 46.64      | 46.64      | 0.989      |
|          | <b>prior+residual (ours)</b>            | 34.95      | 34.95      | 0.898      |
|          | <b>prior+gated residual (ours+gate)</b> | 37.44      | 37.44      | 0.977      |
| 48       | current-only INR                        | 16.61      | 30.44      | 0.936      |
|          | prior-finetune (NeRP)                   | 20.46      | 34.68      | 0.965      |
|          | <b>prior+residual (ours)</b>            | 16.31      | 29.53      | 0.890      |
|          | <b>prior+gated residual (ours+gate)</b> | 16.23      | 30.00      | 0.961      |
| 96       | current-only INR                        | 13.91      | 22.33      | 0.845      |
|          | prior-finetune (NeRP)                   | 19.20      | 27.66      | 0.893      |
|          | <b>prior+residual (ours)</b>            | 16.00      | 24.26      | 0.850      |
|          | <b>prior+gated residual (ours+gate)</b> | 15.93      | 24.29      | 0.911      |
| 144      | current-only INR                        | 12.83      | 17.80      | 0.702      |
|          | prior-finetune (NeRP)                   | 18.63      | 23.61      | 0.782      |
|          | <b>prior+residual (ours)</b>            | 16.20      | 21.14      | 0.789      |
|          | <b>prior+gated residual (ours+gate)</b> | 16.07      | 21.04      | 0.831      |

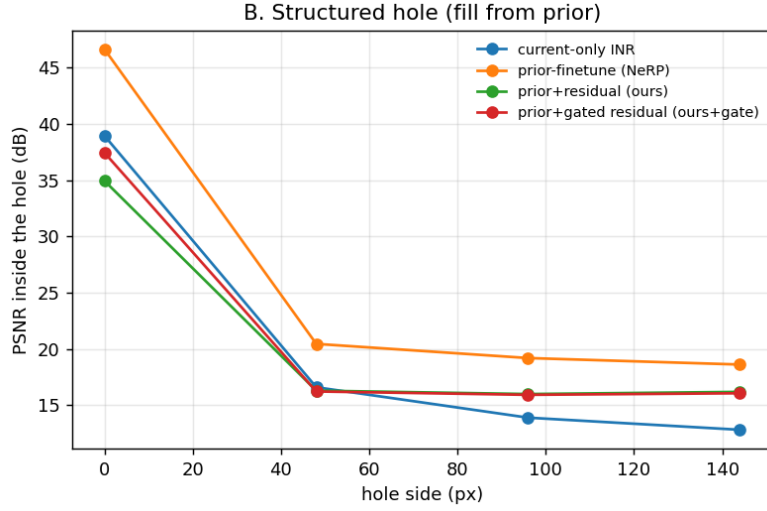


Figure 3: PSNR inside the hole vs. hole size. current-only cannot fill a hole it never saw.

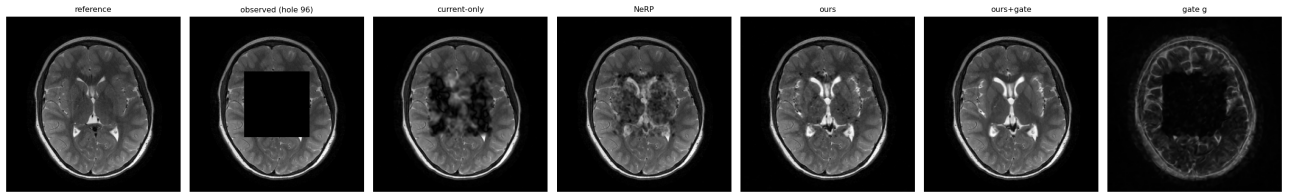


Figure 4: Reconstructions with a 96px hole: reference, observed (hole blanked), then each method, and the gate map.